

Polynomial Basis Constants from the John Ellipsoid

Candidate answer to the exponential-improvement problem in arXiv:2303.03210

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Status. Candidate full answer to the source’s exponential-improvement question, likely valid, subject to human review. The result gives a complete polynomial upper bound, uniformly over the source’s angle parameter, and an exact ellipsoidal subcase. It does not determine the exact worst-case constants in dimensions above two.

1 Source problem

The target is Section 6 (PDF pp. 17–18) of Gerdjikov–Nikolov, *Some inequalities for norms in $\mathbb{R}^n$ and $\mathbb{C}^n$* , arXiv:2303.03210 [1]. Their greedy minimal and maximal bases give the sharp-for-that-strategy constant $2^n - 1$. They ask how this exponential constant can be improved, define the optimal orthonormal-basis constant $c_n^\perp$, the unrestricted unit-basis constant $c_n^\angle$, and the interpolating constants c_n^ρ under $|\langle b_i, b_j \rangle| \leq \rho$.

6. OPEN PROBLEMS

The definitions of f -minimal and f -maximal bases of a norm suggest a simple greedy strategy to find an orthonormal basis $(e_1, \dots, e_n)$ which satisfies the inequality:

$$f\left(\sum_{i=1}^n \alpha_i e_i\right) \geq \frac{1}{2^n - 1} \sum_{i=1}^n |\alpha_i| f(e_i).$$

Of course, the feasibility of this approach depends on the possibility to efficiently solve the optimisation problem:

$$\max_{v \in \mathbb{S}_1} f(v) \text{ or } \min_{v \in \mathbb{S}_1} f(v).$$

Yet, since every (semi)norm f is convex and $\mathbb{S}_1$ is compact, especially the second problem is well studied and efficient methods for its solution stay at hand.

However, the problem with the greedy approach is that the constant $2^n - 1$ grows exponentially with n and it may be inconvenient to prove precise bounds in general. As we have proven in Lemma 6 and Lemma 7 the constant $2^n - 1$ cannot be improved under the suggested greedy strategy. Thus, the natural question that arises is how this constant can be improved while preserving the clear structure of the bases that it implies. In this respect, we consider the following theoretical problems.

First, for a natural number $n \geq 1$, and a linear vector space V with inner product, where $V = \mathbb{C}^n$ or $V = \mathbb{R}^n$. We define $c_n^\perp := c_n^\perp(V)$ to be the least real number such that for every norm $f : V \rightarrow \mathbb{R}^+$ there is an orthonormal basis $(b_1, \dots, b_n)$ such that:

$$f\left(\sum_{i=1}^n \alpha_i b_i\right) \geq \frac{1}{c_n^\perp} \sum_{i=1}^n |\alpha_i| f(b_i) \text{ for all } \alpha_1, \dots, \alpha_n \in \mathbb{F}.$$

It is known that for $V = \mathbb{R}^2$, $c_2^\perp = 2$, [6]. The construction in [6] relies upon defining appropriate areas and the continuity principle to show existence. Is there a more explicit way to define such a basis? To the best knowledge of the authors, the techniques from $n = 2$ do not extend to higher dimensions.

Secondly, for a natural number $n \geq 1$, and a linear vector space V with inner product, where $V = \mathbb{C}^n$ or $V = \mathbb{R}^n$. We define $c_n^\angle = c_n^\angle(V)$ to be the least real number such that for every norm $f : V \rightarrow \mathbb{R}^+$ there is an basis $(b_1, \dots, b_n)$ of unit vectors such that:

$$f\left(\sum_{i=1}^n \alpha_i b_i\right) \geq \frac{1}{c_n^\angle} \sum_{i=1}^n |\alpha_i| f(b_i) \text{ for all } \alpha_1, \dots, \alpha_n \in \mathbb{F}.$$

It is known that for $V = \mathbb{R}^2$, $c_2^\angle = \frac{3}{2}$, [1]. This question is tightly related with John's Theorem [5] which relies on volumes' optimisation.

Figure 1: The open problems in the source paper, PDF p. 17.

Both questions can be uniformly stated as follows. Let $n \geq 1$, and $V = \mathbb{C}^n$ or $V = \mathbb{R}^n$ and $\alpha \in [0, 1]$. Define $c_n^\alpha := c_n^\alpha(V)$ to be the least real number such that for every norm $f : V \rightarrow \mathbb{R}^+$ there is a basis $(b_1, \dots, b_n)$ of unit vectors such that:

$$f\left(\sum_{i=1}^n \alpha_i b_i\right) \geq \frac{1}{c_n^\alpha} \sum_{i=1}^n |\alpha_i| f(b_i) \text{ for all } \alpha_1, \dots, \alpha_n \in \mathbb{F}$$

subject to : $|\langle b_i, b_j \rangle| \leq \alpha$ for all $i \neq j$.

In this framework, $c_n^\perp(V) = c_n^0(V)$ and $c_n^\angle(V) = c_n^1(V)$. We consider that the freedom to vary α may be useful in applications where this kind of inequalities are to be combined with other classical inequalities where the scalar products of the basis' vectors has to be controlled.

REFERENCES

Figure 2: The unified definition of c_n^ρ , continued on PDF p. 18.

We answer the growth question by replacing $2^n - 1$ with n . The basis is explicit once a classical finite-dimensional convex optimization—the John ellipsoid—has been performed.

2 Definitions

Let $V = \mathbb{F}^n$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, with its fixed Euclidean or Hermitian inner product. For a norm f and $0 \leq \rho \leq 1$, let $C_\rho(f)$ be the infimum of all C for which there is a Euclidean-unit basis $(b_1, \dots, b_n)$ satisfying

$$|\langle b_i, b_j \rangle| \leq \rho \quad (i \neq j)$$

and

$$\sum_{i=1}^n |a_i| f(b_i) \leq C f\left(\sum_{i=1}^n a_i b_i\right) \quad (a_1, \dots, a_n \in \mathbb{F}).$$

Then the source's constant is $c_n^\rho(\mathbb{F}^n) = \sup_f C_\rho(f)$, $c_n^0 = c_n^\perp$, and $c_n^1 = c_n^\angle$.

The unit ball of f is denoted by K . An ellipsoid E has a quadratic gauge

$$q(x) = \inf\{t > 0 : x \in tE\} = \|Tx\|_2$$

for a positive definite self-adjoint operator T .

Proof intuition

The exponential loss in the source comes from recursively projecting away one direction and paying a factor close to two at each step. Instead, put the entire norm ball into its best ellipsoidal position at once. John's theorem supplies $E \subset K \subset \sqrt{n}E$. Diagonalizing E uses an orthonormal basis in the original inner product. Along that basis the numerator is a weighted ℓ_1 norm and the ellipsoid is the corresponding weighted ℓ_2 norm. Cauchy–Schwarz costs $\sqrt{n}$, and the John sandwich costs another $\sqrt{n}$, producing n rather than $2^n - 1$. The Euclidean norm forces at least $\sqrt{n}$ by averaging over signs. When the original norm is already ellipsoidal, the sandwich has no distortion and the exact constant is $\sqrt{n}$.

3 The John sandwich over both scalar fields

Lemma 1 (Real and complex symmetric John sandwich). *Let K be the unit ball of a norm on $\mathbb{F}^n$. There is an ellipsoid E , defined by a positive definite self-adjoint $\mathbb{F}$ -linear operator, such that*

$$E \subset K \subset \sqrt{n}E.$$

For $\mathbb{F} = \mathbb{C}$, the ellipsoid can be taken Hermitian relative to the fixed complex structure.

Proof. For $\mathbb{F} = \mathbb{R}$, this is the centrally symmetric form of John's theorem [2]. We recall the contact-point argument because it also resolves the scalar-field issue. After a linear change of coordinates puts the John ellipsoid in the form $E = B_2^n$, there are contact points $u_j \in \partial K \cap S^{n-1}$ and weights $d_j > 0$ with

$$I = \sum_j d_j u_j \otimes u_j, \quad \sum_j d_j = \text{tr } I = n.$$

At a contact point, the supporting functional of K is the Euclidean functional represented by u_j ; central symmetry gives $|\langle x, u_j \rangle| \leq 1$ for $x \in K$. Hence

$$\|x\|_2^2 = \sum_j d_j |\langle x, u_j \rangle|^2 \leq \sum_j d_j = n,$$

which is $K \subset \sqrt{n}B_2^n$.

Now let $\mathbb{F} = \mathbb{C}$ and regard K as a centrally symmetric body in $\mathbb{R}^{2n}$. Its real John ellipsoid is unique. Since K is invariant under every scalar phase $R_t : x \mapsto e^{it}x$, uniqueness gives $R_tE = E$. Thus the positive operator defining E commutes with the complex structure, so E is Hermitian. Normalize again to $E = B_2^n$ and use the real John decomposition

$$I_{\mathbb{R}^{2n}} = \sum_j d_j u_j \otimes_{\mathbb{R}} u_j, \quad \sum_j d_j = 2n.$$

Every $R_t u_j$ is again a contact point. The supporting-functional argument and phase invariance of K imply

$$|\langle x, u_j \rangle_{\mathbb{C}}| \leq 1 \quad (x \in K).$$

Average the real quadratic decomposition over $t \in [0, 2\pi]$. For every $x \in \mathbb{C}^n$,

$$\|x\|_2^2 = \frac{1}{2} \sum_j d_j |\langle x, u_j \rangle_{\mathbb{C}}|^2 \leq \frac{1}{2} \sum_j d_j = n.$$

Thus $K \subset \sqrt{n}B_2^n$ also in the complex case. Undoing the complex-linear normalization proves the claim. $\square$

4 Polynomial upper bounds and matching lower bound

Theorem 1 (Uniform polynomial bound). *For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, $n \geq 1$, and every $0 \leq \rho \leq 1$,*

$$\boxed{\sqrt{n} \leq c_n^\rho(\mathbb{F}^n) \leq n.}$$

More explicitly, for every norm f there is an orthonormal basis $(b_1, \dots, b_n)$ such that

$$\sum_{i=1}^n |a_i| f(b_i) \leq n f\left(\sum_{i=1}^n a_i b_i\right) \quad (a_i \in \mathbb{F}).$$

One may take an eigenbasis of the positive operator defining the John ellipsoid of the f -unit ball.

Proof. Take E from Lemma 1 and write its gauge as $q(x) = \|Tx\|_2$. The inclusions are equivalent to

$$\frac{1}{\sqrt{n}} q(x) \leq f(x) \leq q(x) \quad (x \in V).$$

Choose an orthonormal eigenbasis b_i of T , say $Tb_i = \lambda_i b_i$ with $\lambda_i > 0$. Then $f(b_i) \leq \lambda_i$. If $x = \sum_i a_i b_i$, Cauchy–Schwarz gives

$$\begin{aligned} \sum_i |a_i| f(b_i) &\leq \sum_i |a_i| \lambda_i \\ &\leq \sqrt{n} \left(\sum_i |a_i|^2 \lambda_i^2 \right)^{1/2} = \sqrt{n} q(x) \leq n f(x). \end{aligned}$$

The basis is orthonormal, hence it is admissible for every $\rho \geq 0$.

For the lower bound, use the Euclidean norm $f = \|\cdot\|_2$. Let (b_i) be any admissible unit-vector basis. For independent Rademacher signs ε_i ,

$$\mathbb{E} \left\| \sum_i \varepsilon_i b_i \right\|_2^2 = \sum_i \|b_i\|_2^2 = n.$$

Therefore some sign choice has norm at most $\sqrt{n}$, whereas $\sum_i |\varepsilon_i| f(b_i) = n$. The constant for that basis is at least $\sqrt{n}$. This holds for every admissible basis and every ρ . $\square$

In particular,

$$\sqrt{n} \leq c_n^\perp(\mathbb{F}^n) \leq n, \quad \sqrt{n} \leq c_n^\angle(\mathbb{F}^n) \leq n.$$

For the orthonormal problem this replaces the source's exponential $2^n - 1$ by n . The cited planar identity $c_2^\perp(\mathbb{R}^2) = 2$ shows that the new upper bound is exact when $n = 2$.

5 Exact solution for ellipsoidal norms

Proposition 1. *If $f(x) = \|Ax\|_2$ for a positive definite self-adjoint operator A , then*

$$C_\rho(f) = \sqrt{n} \quad (0 \leq \rho \leq 1)$$

over both scalar fields.

Proof. Let (b_i) be a Euclidean orthonormal eigenbasis, $Ab_i = \lambda_i b_i$. It is admissible for every ρ , and

$$\sup_{a \neq 0} \frac{\sum_i |a_i| f(b_i)}{f(\sum_i a_i b_i)} = \sup_{a \neq 0} \frac{\sum_i |a_i| \lambda_i}{(\sum_i |a_i|^2 \lambda_i^2)^{1/2}} = \sqrt{n}.$$

Equality is attained by $|a_i| = 1/\lambda_i$.

Conversely, take any unit-vector basis (v_i) and put $x_i = v_i/f(v_i)$, so $f(x_i) = 1$. Independent signs satisfy

$$\mathbb{E} f\left(\sum_i \varepsilon_i x_i\right)^2 = \mathbb{E} \left\| \sum_i \varepsilon_i A x_i \right\|_2^2 = \sum_i \|A x_i\|_2^2 = n.$$

This rescaling leaves the coefficient ratio unchanged: choose $a_i = \varepsilon_i/f(v_i)$ in the original basis. For some sign choice the denominator is at most $\sqrt{n}$, while the weighted coefficient numerator is n . Thus no basis has constant below $\sqrt{n}$. $\square$

6 Verification, novelty, and limitations

The script `code/check_bounds.py` checks the exact ellipsoidal witness, exhausts all signs for random real and complex bases through dimension eight, and reports 28,032 deterministic or seeded-random sign, ellipsoid, and coefficient checks, including synthetic norms satisfying the John sandwich. Run from the packet directory with

```
conda run --no-capture-output -n sandbox python code/check_bounds.py.
```

These computations are sanity checks only; the proof is the preceding ellipsoid argument.

Novelty search. On 17 August 2026, bounded searches covered the run’s registry, solution, attempt, and proof-gap indexes; the exact title and arXiv id; the symbols and phrases for $c_n^\perp$, $c_n^\angle$, and c_n^ρ ; and arXiv searches combining John ellipsoids, orthonormal bases, weighted coefficient inequalities, and crosspolytope containment. No later paper explicitly answering the source problem was located. The result is an elementary corollary of classical John theory, so novelty confidence is moderate rather than high.

Limitations. The theorem fully answers the source’s request to improve the exponential constant by giving a uniform linear bound and a concrete basis-selection principle. It does not determine the exact values of c_n^ρ in general for $n > 2$, nor does it prove that the upper bound n is asymptotically sharp. John ellipsoid computation is a standard convex optimization, but the packet does not analyze its numerical complexity or give a closed formula for an arbitrary input norm.

Human-review recommendation. Verify the complex part of Lemma 1, especially phase averaging of the real contact decomposition, and the gauge direction in $E \subset K \subset \sqrt{n}E$. Then assess whether the durable result should be described as a full answer to the exponential-growth question or as a substantial partial answer to the broader exact-constant problem.

References

- [1] S. Gerdjikov and N. Nikolov, *Some inequalities for norms in $\mathbb{R}^n$ and $\mathbb{C}^n$* , arXiv:2303.03210v1 (2023), Section 6.
- [2] F. John, *Extremum problems with inequalities as subsidiary conditions*, in *Studies and Essays Presented to R. Courant on His 60th Birthday*, Interscience, 1948, pp. 187–204.